Lesson 07 Demo 04

Integrating React Libraries: Axios and Formik

Objective: To create a React application that utilizes Formik for form handling and validation and Axios for remote data submission

Tools Required: Visual Studio Code

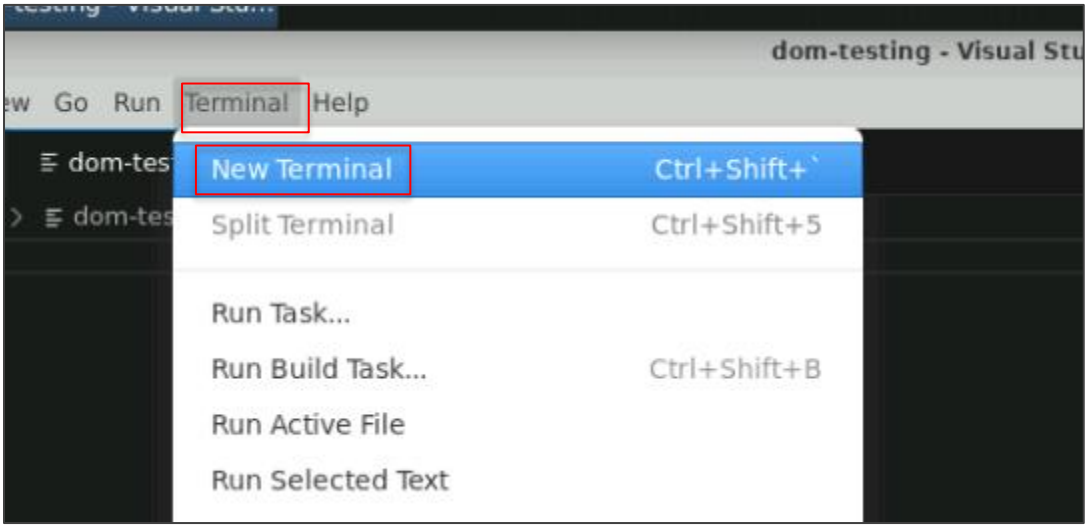
Prerequisites: Knowledge of creating a React app and understanding of the folder structure

Steps to be followed:

1. Set up a new React project using Vite
2. Import and use the component
3. Run and verify the application

Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code, click on **Terminal,** and then select **New Terminal**



1.2 Open the terminal and run the following command:

npm create vite@latest react-libraries-demo

```
labuser@ip-172-31-30-52:~$ npm create vite@latest react-libraries-demo

> npx
> create-vite react-libraries-demo
```

1.3 Select **React** and press enter

```
Select a framework:
  Vanilla
  Vue
  React
  Preact
  Lit
  Svelte
  Solid
  Qwik
  Angular
  Others
```

1.4 Select **JavaScript** and press enter

```
Select a framework:
React

Select a variant:
  TypeScript
  TypeScript + SWC
  JavaScript
  JavaScript + SWC
  React Router v7 ↗
```

1.5 Run the following command to navigate to the project folder:

cd react-libraries-demo/

```
◇  Select a framework:
● labuser@ip-172-31-30-52:~$ cd react-libraries-demo/
● labuser@ip-172-31-30-52:~/react-libraries-demo$ npm i
```

1.6 Install Axios and Formik using the command given below:

npm install axios formik

```
^C
○ labuser@ip-172-31-30-52:~/react-libraries-demo$ npm install axios formik
```

1.7 Run the following command to install dependencies:

npm i

```
● labuser@ip-172-31-30-52:~/react-libraries-demo$ npm i

  added 149 packages, and audited 150 packages in 21s

  30 packages are looking for funding
    run `npm fund` for details

  found 0 vulnerabilities
```

1.8 Run the following command to start the development server:

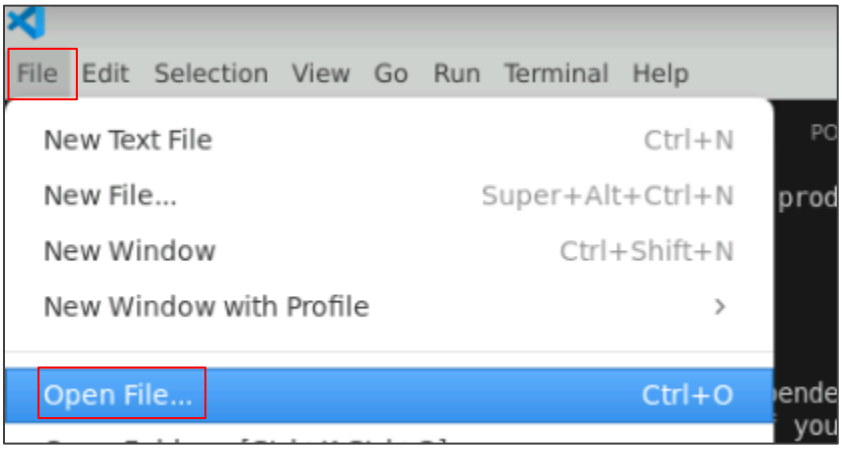
npm run dev

```
○ labuser@ip-172-31-30-52:~/react-libraries-demo$ npm run dev

  > react-libraries-demo@0.0.0 dev
  > vite


    VITE v6.2.2  ready in 402 ms
```

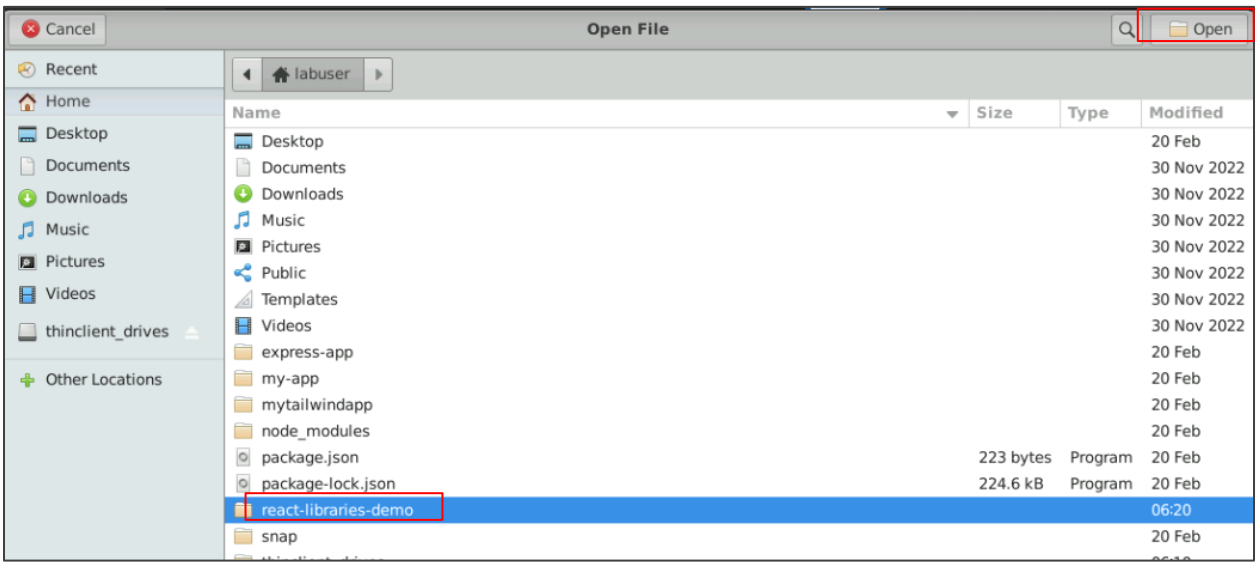
1.9 Navigate to the web browser and paste the link given below:
http://localhost:5173

Step 2: Import and use the component

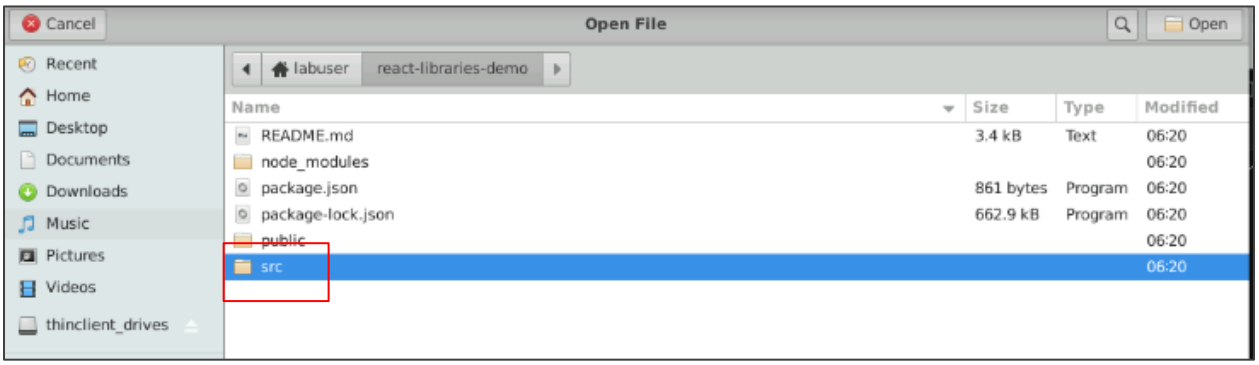
2.1 Open Visual Studio Code, click on **File**, and select **Open File…**



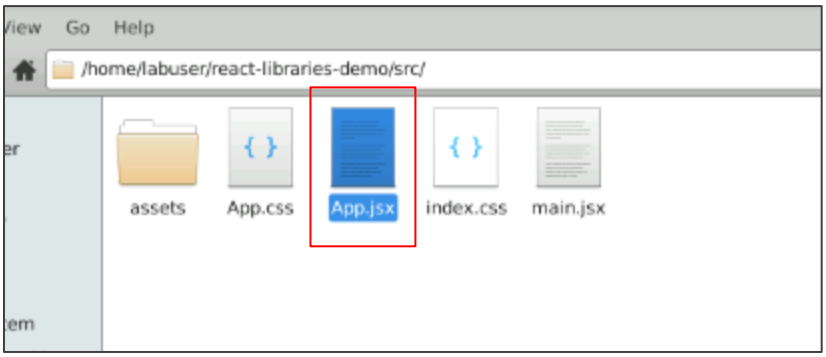
2.2 Select **react-libraries-demo** and click **Open**



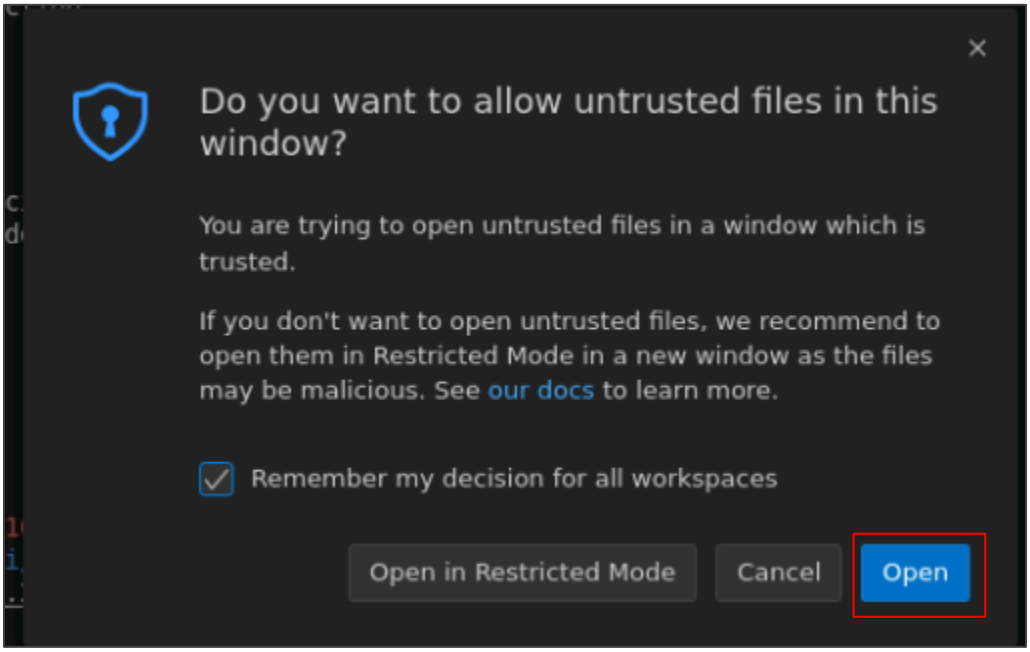
2.3 Double-click **src**



2.4 Open **App.jsx**



2.5 Click **Open**



2.6 Modify **App.jsx** and add the following code:

```
import React from 'react';
import { Formik, Form, Field, ErrorMessage } from 'formik';
import axios from 'axios';

function App() {
  const handleSubmit = (values, { setSubmitting }) => {
    axios.post('https://jsonplaceholder.typicode.com/posts', values)
      .then(response => {
        console.log(response);
        setSubmitting(false);
      })
      .catch(error => {
        console.log(error);
        setSubmitting(false);
      });
  };

  return (
    <div>
      <h1>React Libraries Demo</h1>
      <Formik
```

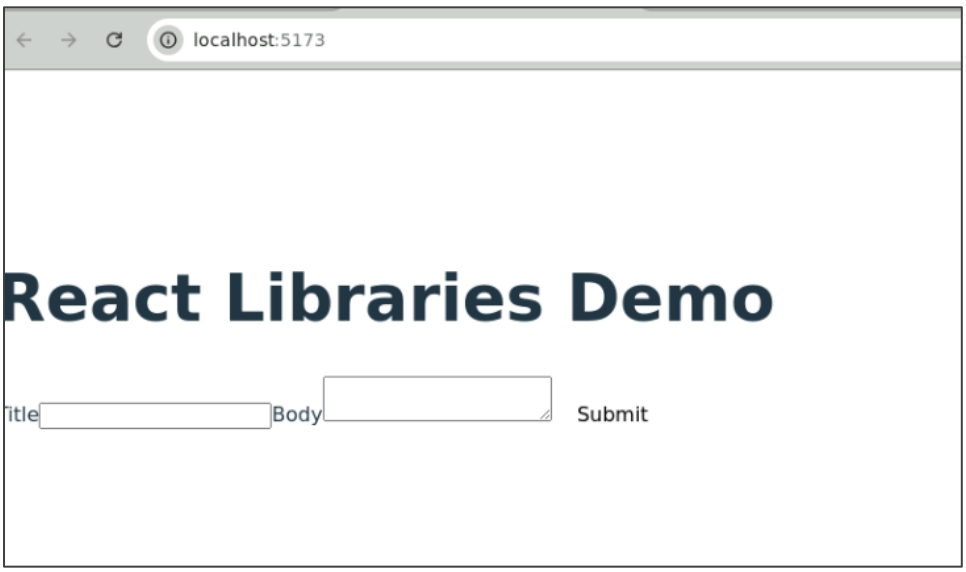
```
            initialValues={{ title: '', body: '' }}
            validate={values => {
              const errors = {};
              if (!values.title) {
                errors.title = 'Title is required';
              }
              if (!values.body) {
                errors.body = 'Body is required';
              }
              return errors;
            }}
            onSubmit={handleSubmit}
          >
            {({ isSubmitting }) => (
              <Form>
                <label htmlFor="title">Title</label>
                <Field type="text" name="title" />
                <ErrorMessage name="title" component="div" />

                <label htmlFor="body">Body</label>
                <Field as="textarea" name="body" />
                <ErrorMessage name="body" component="div" />

                <button type="submit" disabled={isSubmitting}>Submit</button>
              </Form>
            )}
          </Formik>
        </div>
      );
    }

    export default App;
```

```
home > labuser > react-libraries-demo > src > JS App.js > App
1
2   import React from 'react';
3   import { Formik, Form, Field, ErrorMessage } from 'formik';
4   import axios from 'axios';
5
6   function App() {
7       const handleSubmit = (values, { setSubmitting }) => {
8           axios.post('https://jsonplaceholder.typicode.com/posts', values)
9               .then(response => {
10                  console.log(response);
11                  setSubmitting(false);
12              })
13              .catch(error => {
14                  console.log(error);
15                  setSubmitting(false);
```

Note: App.jsx uses two libraries, **Axios** and **Formik**. These libraries help build a user-friendly and functional form in React that handles validation and efficiently sends data to a backend server.

Step 3: Run and verify the application

3.1 Navigate to the browser and refresh the web page



By following these steps, you have successfully created a React application that demonstrates the usage of Formik for form validation and Axios for remote data submission. This setup allows efficient handling of form state and API interactions within React applications.